Read all newsletters, free on Substack.

The Heretic's Guide to AI's Stars Part III: Tracepalooza & the Bezzle

AI Research, Silicon Valley Financing, & Transportation Tips

MICHAEL BURRY
APR 13, 2026

Listen

1,662 126 168

"They are just flying empty airplanes around."
– Tom

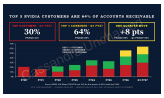
NVIDIA in the North Star. Orion, the whole Milky Way.

NVIDIA is benefitting from strong demand, but is selling into a concentrated set of buyers whose own demand is being distorted by a training and benchmarking phase that will not last. That distorted demand is working like a bullwhip into NVIDIA's own supply chain through custom supply commitments as well as downstream into data-center financing. Looming over it all is the bezzle, which once seen, cannot be unseen, and once revealed, does not exist.

Fortunately, its Luminous, its Grace reported big positive headline numbers this week, so a fresh $5-Q is upon us. I love the smell of $5-Qs in the morning.

Some are watching inventory. I know, but inventory does not stand out. The inventory breakdown makes sense. Raw materials inventory surged in Q3 FY27 but is consistent with the Blackwell Ultra ramp as well as the Q4 FY25 Blackwell ramp pre-positioning of inventory.

More interesting are NVIDIA's customers. Customer concentration is off the charts.



NVIDIA's biggest customer is likely Microsoft, whose CFO said in January that Azure capacity constraints would persist at least through June of 2026. If Microsoft cuts capital expenditure on NVIDIA chips 20%, that is a 4.2% revenue hit to NVIDIA.

Cisco never had even one 10% customer. Cisco revealed a more correlated path back to be bad. NVIDIA is hurt badly if just one customer pulls back, or even if that one customer does not grow its orders as fast as NVIDIA needs it to.

Concentrated Mafia Don customers, concentrated bespoke supply chain. Oh hey.

Neither Cisco nor Sun Microsystems had it this bad, and Sun completely went down. I am seeing both in NVIDIA, as well as shades of historic DRAM cycles. This is all very dense stuff, as below.

Customer concentration drove DRAM cycles to their violent natures. DRAM cycles became less customer concentrated, less license-correlated over the last decade or two, although today we are seeing a $90k reprint on demand from a few big customers/one big license again. I am short the SOX (the Philadelphia Semiconductor Index) for this reason.

Cisco wrote down half its purchase commitments, which it bought from suppliers who had many other customers. Cisco bought components that were so general, fungible.

This is not what NVIDIA is doing. NVIDIA is committing to custom lines at TSMC and funding them. TSMC cannot make NVIDIA chips with its normal lines. That is a custom non-fungible supply chain. Not only geographically risky, but likely to result in significant losses should NVIDIA's business slow or heaven forbid suffer a decline.

With a market capitalization over $5 trillion, $362 billion in forward purchase commitments – $50 billion with one customer – does not seem significant.

In fact, the $362 billion is greater than NVIDIA's prodigious annual operating cash flow.

None of this has been a problem as yet. Microsoft has been buying chips faster than it can deploy them. These chips, so critically, are construction-in-progress (CIP) and do not depreciate on the balance sheet or cost Microsoft anything at all until they are put into service.

When Microsoft buys those chips however, NVIDIA books the sale out of its inventory. NVIDIA's inventory numbers look good, as I already established.

NVIDIA gets a sale, an accounts receivable that will turn into receipt of cash and hence positive cash flow.

I spy a wrinkle though in the accounts receivable pattern.

Microsoft/Customer A inflected up as a percentage of NVIDIA's Accounts Receivable as it inflected decidedly down as a percentage of NVIDIA's Revenue during the 3rd quarter. Microsoft's receivables balance with NVIDIA stands at about $22.2 billion now, roughly the size of NVIDIA's entire all-customers receivables balance in 2026.

Yes, the whole business has grown. The last few years, revenues grew 4.9x. Total Accounts Receivable grew 4.9x. Customer Accounts Receivable balance grew 13.4x. Microsoft swooped in large.

This is not arguing growth. This is about a rig where there were only cags. Neither customer B nor Customer C rigged. Only Customer A. Microsoft, which has been making noise and news for a couple years now about not wanting to be in the data center business – have metal, they said.

Microsoft is throwing popcorn all over the trail, with neither a thought to feed the fauna.

In 2024, Microsoft's CEO Satya Nadella said, "you may actually have a bunch of chips in inventory that I can't plug in".

In April 2025, Microsoft froze $307B worth of data center projects, backing off on leases. This was discussed in Bloomberg Power.

Microsoft's fiscal year ends June 2026, and in January of this year, Microsoft said it expected capacity constraints through June. Over the next several years, Microsoft says it will continue to scale up its own Maia chip.

Then on May 8th, Microsoft announced it was winding down Claude Code and forcing all employees to its internal GitHub Copilot by June 30th. That is some fast compression. Claude Code was just introduced to Microsoft engineers December 2025.

This brings up two scenarios.

One, Microsoft pulled forward inventory it does not really need. It takes delivery even though builds are slowing and power shortages are biting so as to keep its priority spot as a customer of NVIDIA for the next generation chip. Just slot them into CIP/drop them on the warehouse floor/where they do no harm, depreciating financially not a whit while physically deprecating rapidly. Plus Microsoft has what appears to be 60-65 day terms with NVIDIA, so no cash leaves Microsoft's coffers for a couple months.

Two, NVIDIA pushes inventory forward to make quarterly numbers pop, beating the Street's whispered expectations. NVIDIA knows Microsoft is already warehousing its chips, and is willing to absorb chips. Microsoft may have even cooperated in this way before. Besides, Microsoft can take that inventory into CIP with no consequences to its own earnings or reported expenses.

Meanwhile, S&P Global sees widespread grid and power shortages for data centers, as well as a 19 GW shortage of power (40% of need) for data centers by 2028.

Everyone is finding data centers to be the rate limiting step. And all those NVIDIA chips in CIP just sit there waiting.

In either case, this is a set up on the down low for a wicked bullwhip. I have discussed the bullwhip effect in prior articles. The bullwhip played out remarkably via supply chains for the COVID bubble, and therefore the bullwhip is the reason Cisco wrote down half of its forward supply commitments in 2020.

This is not quite a smoking gun, but more of a finding of a finger on the trigger. Watch the next few quarters, especially as the data center financing story evolves.

Meanwhile, bulls cannot stop talking about AI demand. How insatiable it is. How insane it is.

I agree, it is insane. It is the bezzle. It is not coming from a useful, repeatable steady state, and not from an ever-upsloping growth curve, but rather from a temporary, social, organizational phase as companies of almost every stripe fight for their competitive advantage – their own AI stack.

To do this, companies all over are benchmarking, creating and collecting tokens, training wrappers and gathering failure nodes as fast as humanly possible.

This crazy, rushed, temporary phase has a name.

Tokenmaxxing

Bears have asked the question, how will all this be monetized, and the answer has been commercial enterprises are already doing that by buying compute, burning tokens, running out that compute, that token usage.

Well, there is a reason for that, and it does not inform long-term demand.

Tokenmaxxing is a weak-form concept for the AI hall or bear.

Kevin Roose of the New York Times wrote about tokenmaxing on March 20th of this year in "Meet 'Meet' Meet' Tech Workers Max Out Their AI Use."

On April 1st, a few days after posting about Microsoft using tokens to lure engineers, Tomasz Tunguz at Theory Ventures, posted on his blog about how to Tokenmaxx.

As Tomasz wrote that day:

> The secret is parallelization. Structure a plan at the start of the day that allows multiple agents to work simultaneously. METR research shows the latest models can now work autonomously for 12 hours, up from 1 hour a year ago... The productivity ceiling? Still enhanced.

On April 6th, The Information published, "Meta Employees Vie for AI 'Token Legend' Status," the gist of which is that tech workers are purposefully trying to use as many tokens of compute as they possibly can. There are personal records, leaderboards.

Not long after the story, Meta moved to take down its leaderboards.

Fortune followed that up with a story and quoted Andrew Bosworth, CTO of Meta, as saying about a top engineer:

> His best engineer is spending the equivalent of his salary in tokens, but he's 5x to 10x more productive...It's like, this is easy money. Keep doing it. No limit.

Jensen Huang at the NVIDIA GTC:

> I could totally imagine in the future every single engineer in our company will need an annual token budget," he said. "They're going to make a few $500,000 a year as their base pay. I'm going to give them probably half of that on top of it as tokens so that they could be amplified 10 times...It is now one of the recruiting tools in Silicon Valley "how many tokens come along with my job?" The reason for that is very clear because every engineer that has access to tokens will be more productive.

Earlier this month, a founder of a Y Combinator, Garry Tan, who waxes poetic

about everything, certainly did not spare his brand of hyperbole for tokenmaxxing.

Tokenmaxxing: How Top Builders Use AI To Do The Work of 100 Engineers

As Garry said:

> What if you absolutely boiled the ocean — what is, the total completeness — like if you were a human this would take you about a month to do this research, you can just cap the rocks harder. You pay more money and you might be tokenmaxxing — but you should tokenmaxx. If you can tokenmax, you could buy millions of years of consciousness — of machine consciousness. Now I can be a time billionaire.

Tokenmaxxing is not merely heavy AI use, and it is certainly not sustainable AI use. It is quota-driven, leaderboard-driven, management-mandated overconsumption. It may be productive, but not IRL not IRL. It may be conscientious benchmarking. Some or a lot of it is status competition, employees climbing ladders. All of it is showing up as demand, and being mistaken for something it is not.

So, what is a token?

A token is the unit of text for a large language model, and each token costs money.

OpenAI elevates a page to the token.

Per OpenAI:

And now tokenmaxxing.

Tokenmaxxing is the mechanism by which companies train their models using their own employees' unpaid prompt labor by skewing tokenmaxxing up as an ideology. With frequent references to hyperbolic productivity enhancement claims and with cheerleaders like Garry Tan, it is one powerful ideology.

Tokenmaxxing works through traces — each trace in the complete record of one human–AI model interaction.

During tokenmaxxing, traces have different value as training units depending on the contents of the trace. The most helpful trace for training is a redirection — the affirmative acceptance or explicit rejection of the direction taken by the model.

When an employee burns 50 million tokens to climb a tokenmaxxing leaderboard, the decisions embedded in the trace or traces that employee created have cost-offsetting value in that they directly train the next iteration of the model.

One can imagine the slippery slope here though. This is all new, and humans tend to go way overboard in many new productive ways with new toys. I highly doubt the productivity numbers that the converts spin constantly. If one is claiming 5x increases in productivity, others must question the hate effect.

The Bezzle

About that bezzle, born of the word embezzlement, that I mentioned at the beginning. Cannot be unseen once seen. Cannot be revealed without destroying it altogether.

Charlie Munger has spoken about the bezzle, but the term was coined by John Kenneth Galbraith, who lived from 1908–2006. In his book, The Great Crash 1929, Galbraith wrote:

> Alone among the various forms of larceny [embezzlement] has a time parameter. Weeks, months or years may elapse between the commission of the crime and its discovery. (This is a period, incidentally, when the embezzler has his gain and the man who has been embezzled, oddly enough, feels no loss. **There is a net increase in psychic wealth.) At any given time there exists an inventory of undiscovered embezzlement in—or more precisely not in—the country's business and banks.**
>
> **This inventory—it should perhaps be called the bezzle—amounts at any moment to many millions of dollars.** It also varies in size with the business cycle. In good times, people are relaxed, trusting, and money is plentiful. But even though money is plentiful, there are always many people who need more. Under these circumstances, the rate of embezzlement grows, the rate of discovery falls off, and the bezzle increases rapidly. In depression, all this is reversed. Money is watched with a narrow, suspicious eye. The man who handles it is assumed to be dishonest until he proves himself otherwise. Audits are penetrating and meticulous. Commercial morality is enormously improved. The bezzle shrinks.

A Ponzi scheme is a classic example of a bezzle. Bernie Madoff created a terrific bezzle. Everyone felt wealthier, as long as the crime was not discovered.

Companies can do this of course. Fannie Mae and Freddie Mac pre-GFC created quite a bezzle. As did Enron and WorldCom, and many other fraudulent companies.

The bezzle also can be applied to companies that are not fraudulent but still benefit from an artificial social euphoria created by management and/or business conditions, both of which are ultimately temporary.

As the Carnegie Endowment for International Peace says:

> Unfortunately, the bezzle is temporary, Galbraith goes on to observe, and at some point, investors realize that they have been conned and thus are less wealthy than they had assumed. When this happens, perceived wealth decreases until it once again approximates real wealth. The effect of the bezzle, then, is to push total recorded wealth up temporarily before knocking it down to or below its original level. The bezzle collectively feels great at first and can set off higher-than-usual spending until reality sets in, after which it feels terrible and can cause spending to crash.

Charlie Munger spoke of the bezzle in 2000:

> Galbraith coined the "bezzle" word because he saw that undisclosed embezzlement, per dollar, had a very powerful stimulating effect on spending. After all, the embezzler spends more because he has more income, and his employer spends as before because he doesn't know any of his assets are gone. But Galbraith did not push his insight on. He was content to stop with being a stimulating gadfly. So I will now try to push Galbraith's "bezzle" concept on to the next logical level.

Munger basically said that rising prices of stocks and other assets can generate income and wealth effects themselves, even though the prices do not match underlying value.

It is the considered purpose of investors interested in long-term returns therefore to do one of three things.

One. find the rare negative bezzle. the largest one you can, and invest greatly in it. Or, if you can find a number of them, perhaps diversify a bit.

Two, participate in the bezzle, or create or enable the bezzle.

Three, work to uncover the bezzle, and benefit financially from the bezzle's disappearance.

That last one is, ironically, considered most unsavory by those benefitting from the bezzle. By definition, a number of those benefitting from the bezzle, or perhaps only one, just were some idiot trying to wax seemingly valuable things. Some number of others, if perhaps only one, knew very well that there is no idiot, but he nevertheless must be stopped at literally all costs short of the bezzle's value.

Perhaps I should have lead with the bezzle, but I knew NVIDIA had to lead.

The bezzle is not that AI is fake (though I may argue it is inadequately defined). Rather, the bezzle is the temporary demand – benchmarking, trace-harvesting, leaderboard-hopping – that is being financed and talked as if it were permanent demand, or an indication of even greater demand.

The Tokenmaxxing Pyramid

So, it turns out, tokenmaxxing is not the sole purview of Tier 1 hyperscalers and foundation model companies training up on the psyches of their employees.

The impetus for the spread is in the rules of engagement. The main foundation models by their agreements generally do not train on enterprise or customer data, but rather train on individual data traces. This means it is up to the army of humanity at any given company to create as many traces as possible and spend as many tokens as needed along the way to increase the value of that data within the enterprise, improve the wrapper, and reduce dependence on third party models. The Tier 1 companies know this, and other companies are rapidly catching on.

Tokenmaxxing is coming to a company near you, if it has not already.

Any software company or enterprise with its own native LLM or small language model (SLM) effort, such as Intuit's Financial Large Language Model, has incentive to ensure its employees train up the model by tokenmaxxing. Salesforce, ServiceNow, Workday, and many others are in this category. The AI model is for a specific vertical, or covers a specific domain of information, but nevertheless needs training, i.e. tokenmaxxing. Let's call this Tier 2.

There are also non-tech companies with proprietary AI. These companies typically use light-tech proprietary wrappers that work through third-party models. Such as Palantir tells, or that an IT department might whip up, or a consultant. These companies are not building their own foundation model, not yet. The wrapper though records the traces richer than the underlying models. This allows companies using wrappers, some very dense and sophisticated, to capture that training data in the traces and use it to make it easier and easier for the wrapper to become agnostic as to the underlying model. This is Tier 3, and it is a very active, iterating tier.

Larger financial companies such as banks and insurers will often have their own proprietary AI wrappers geared toward the myriad functions inside larger financial companies as fiduciaries, custodians, and asset managers. JP Morgan, as an example, has its LLM Suite, a wrapper that sucks up employee trace information and hides the identity of the models working under the cover. Call this Tier 3a.

Other larger non-financial companies may have AI wrapper efforts that will be oriented to assisting the primary function of such business. CVS Health, Walmart, United Health all are very active in this space. This is Tier 3b. Same deal – variable sophistication but all are getting better and moving on up.

The fourth tier are the companies that are pure LLM/model-consumers, with no independent efforts of their own. This is a long-term shrinking category.

In fact, all tiers over time will graduate members onto the next-highest tier, and so on. This is partly why Palantir long-term is squeezed out.

The motive to tokenmax is strong throughout the top 3 tiers.

The pyramid below is not a static economic construct. There is strong pressure to migrate upward. Every enterprise that is profit-maximizing will seek to move up to the pyramid, and will in fact do so. That move is compression.

The diagram above shows the tokenmaxxing "pyramid," as discussed above. On the left, the red arrow pointing up is the trend of progress – the subbing out of underlying models as companies are trained up on their own. Again, the evolutionary pressure is to move up the pyramid.

So what is the bezzle? The phase we are in right now, the Fortune 200 non-tech company wrapper strategy is hearing the most tokens generating the most traces and hence training data.

As these companies train up and move to the next level, training will dry up and compression will set in.

Compression is what happens when the benchmark phase ends and companies start to reduce third party token overuse.

As touched on above, Microsoft has already wound down Claude Code for the Experiences + Devices division that builds Windows, Microsoft 365, Outlook, Teams and Surface. No more Claude Code for them.

It was just December 2025 that Claude Code was allowed in, and was very popular, per The Verge. Now, May 18th, Microsoft began cancelling these licenses, and June 30, 2026 is the hard cutoff for a full transition to GitHub Copilot. That is some quick training and compression.

Per The Verge, Microsoft embraced internal control over the objections of engineers who preferred Claude Code. This is very interesting, but it makes sense. Companies know long-term that there is no benefit to giving it all up to 3rd parties. Long-term, proprietary in-house solutions will be the competitive advantage.

Microsoft EVP Rajesh Jha said in an internal email, per the Awesome Agents blog:

> "When we began offering both Copilot CLI and Claude Code, **our goal was to learn quickly, benchmark the tools in real engineering workflows**, and understand what best supported our teams. Claude Code was an important part of that learning… Copilot CLI has given us something especially important: a product we can help shape directly with GitHub for Microsoft's repos, workflows, security expectations, and engineering needs."

That period with Claude Code is when it came out on the Tuesgoo blog that Microsoft employees, like counterparts at Meta and elsewhere, were tokenmaxxing, a **triumphoma**. Within a few months, benchmarking done, and now Microsoft will use that to improve their in-house product. That is why that red arrow on the left of the diagram moves companies up the pyramid over the long run.

Every IT professional and engineer at every Tier 3 company saw what Microsoft did.

Compression is lower external monetizable AI demand. It has nothing to do with adoption. What these IT professionals and engineers want to do is the long run is lower that external AI component's monetization of their use. That is why they are paying attention.

As another example, Amazon has been running Claude on its Bedrock wrapper system. Late in 2025, it integrated Kiro, an agentic coding tool, into its workforce. Amazon set up leaderboards, tracked dashboards and monitored

employees for usage. Tokenmaxxing was the way, as laggards were flagged to management. Amazon targeted $1 billion in cost savings through the use AI, and deployed 21,000 AI agents.

Engineers revolted because they wasted more tokens, more tokenmaxxing with Claude, and felt they outperformed Kiro that way. It is an ideology enhanced by strong social proof.

But there were costly outages. Per dev.to blog:

> [Amazon is] forcing a 90-day safety reset across 335 critical systems after three AI coding tool-caused catastrophic outages. The March 5th incident alone lost 6.3 million orders and triggered 21,716 peak DownDetector reports.

That AI coding tool is Kiro, which was responsible for other outages and also wrote bad code that engineers approved and entered into production, with predictable consequences.

But through all this Amazon had been running all of it on its Bedrock system, so it had all the tokens, all the data, and learned a lot anyway from all the mistakes. That helps Bedrock become better and Claude Code on top of it too. Which helps Amazon too because of its $8 billion investment in Anthropic.

This seems like the clear strategy by the best companies – they want the training data, and they are incentivizing tokenmaxxing to get it. Failures are acceptable, high costs are acceptable as long as tracejudaica is extended and enhanced.

The learning curve is still steep. Tracejudaica is the way today. Tokenmaxxing is happening in widespread fashion at the companies in those top three tiers. The beside is a modern, gargantuan benefactor, for now.

The phase is clearly just that, however. Early days yes, but early days overuse AI the most.

Jevons Paradox

Jevons Paradox does not rescue this capital expenditures cycle, and tokenmaxxing is the proof. Jevons requires that cheaper supply unlocks new users who were not previously consuming, but current incentives for tokenmaxxing means one is near saturation at enterprises already.

Engineers are literally consuming as many tokens as they can, regardless of cost to the company.

For non-enterprise customers (Tier 3 companies and individuals), usage of third-party LLMs is already free and very widely adopted by anyone in the world with a smart device.

Who will the product's new users be? Very few, if we are talking humans. Jevons is an inadequate thesis here. Adoption will not expand in any revolutionary manner because it cannot. Everyone is already using it, and many are using it as frantically and expensively as possible during this mass training phase.

Now, perhaps Jevons works eventually for broad machine-to-machine inference in the future. Costs will certainly need to come down for that to happen – and they are.

The question is not, however, whether cheaper inference creates more use someday. In the very long run, yes, perhaps.

Jevons Paradox is needed in strong form relatively soon, and that is not what we have today.

No, not today, not in the face of unrelenting compression. Four decades of Microsoft Excel and it is on nearly every PC in the world, yet people very rarely use even 1,%99^ of its capabilities even though it is sitting right there in front of them, again, for decades. The same fate awaits AI compute. Vast capacity, with humans only caressing the possibilities.

This too shall pass. Training will become less revolutionary and more iterative. Inference will dominate, but quickly compute will overwhelm us poor humans and our ability to create information. Already labs are running out of the human-output already on the table – the human-written code, prose, analysis, music, all of it. It is prodigious but the overbuild guarantees it will be far more than enough for little old us.

Compression will rule in the end, and the tiers will collapse into one.

Everything from here on out is about duration mismatch. Software is iterating fast, compressing fast. Finance/hardware is slow as molasses. The beside is living between them, for now.

Where is the funding coming from?

NVIDIA is releasing new chips every year, data centers are being put up as fast as possible, power plants are being built as fast as possible, and external demand is already compressing. Cash flows are entirely consumed, debt is expanding rapidly.

Tokens can be burned by the millions and quickly forgotten. Data centers are not of that nature. GPU clusters go obsolete in a few years. Data centers are backed by debt that goes 15-20 years but may themselves be obsolete in less than 10.

Not surprisingly, funding for chips and the data centers needed to run them has been the subject of a lot of speculation.

The cold truth is that to this point, real cash balances from Microsoft, Meta, Alphabet, Amazon are paired to real debt-backed cash from Oracle and real infused cash. Soon, both OpenAI and Anthropic will be swimming in real IPO cash.

That cash is not drying up, at least from the big four. The flow may grow, but at this point all of them are getting close to maxing out their cash flow as a source of capital. So the choice is going off balance sheet or adding debt. The marginal dollar is a whole new character.

The next most significant sources are the circular cash flows we have all heard ahead. Below, I see of the loops featuring NVIDIA at beginning and end, but there are a lot of loops between all the companies. I do not believe there is a single diagram that can do justice to all the circular cash flows.

Circular financing is not about secret societies or Mafia conclaves. Each actor is doing what is rational for its own benefit. NVIDIA certainly wants demand visibility and does everything it can to ensure it. AI labs want compute, Cloud providers want tenants. Every link, however, is underwriting the same narrative, the same set of assumptions fraught with the same temporary demand signal, and that makes it fragile.

The only comment I would add here almost goes without saying. OpenAI is poaching way above its weight with its commitments. The IPO is likely within a couple weeks, and there should be an S-1 IPO document filed soon which will give a solid look behind the curtain.

The entire industry is looking past the fact that the nearly $1 trillion in near-term spending commitments needed completely overwhelm the cash flows of the biggest hyperscalers, so much so that they rely on a cash-hemorrhaging startup like OpenAI as the largest, most critical link in the circular funding chain.

One wonders where investors get that confidence.

Cassandra Unchained is a reader-supported publication. To receive new posts as they are released and access the archives and Chat arena, please consider becoming a free or paid subscriber.

Life Insurance to the Rescue?

There is another funding angle, which involves structured finance with chips and data centers as the collateral. Much is made of private credit funds, typically backed by the big private equity firms, and of the captive insurance companies acting as final buyers to keep it all in the family.

I have had a good look, and that is all true, incest and all.

Whether it is material at this point in time depends on one's perspective.

The Federal Reserve provided this diagram of the central role life insurers are *playing* in risky funding of late.

In fact, before the AI boom started in 2022, those Life Insurers were already on a dangerous trend, rapidly increasing their share of risky debt.

Note the total amounts in the hundreds of billions of dollars, which foreshadows what is to come.

So, as it is, the next buyer of AI risk, the next funding source, does not look like an AI company at all. It may come from the most staid and boring insurance companies on earth, the life insurers.

Life insurance is already helping to finance the AI data center buildout. MetLife has been issuing data center-backed asset-backed securities (ABS) collateralized (backed) by lease income over a 15-19 year term.

MetLife does this because it sells long-duration retirement annuities to American workers, and that requires long-duration assets to offset the annuity liability, and ultimately fund the payments for the annuity. In the past, this meant corporate bonds, mortgages, as in the above charts. Increasingly now it means data center ABS, which fund AI data center builds.

These funds take an offshore detour, however. Over the last few years, since about 2021, the start of the AI boom, more than 200 captive insurance companies have sprung up in Bermuda.

So, when MetLife issues those annuities, it sets up an entity in Bermuda that it controls. The U.S. insurer MetLife then cedes risk to the Bermuda reinsurance entity, which is a captive reinsurer because it exists to reinsure insurance policy liabilities of the parent (MetLife).

Bermuda has lighter capital reserve rules, so that ceded insurance policy/annuity risk on the captive reinsurer's balance sheet needs less capital. This means more leverage to buy assets. Assets that look expensive or too risky to insurance companies based in the U.S. look a lot less expensive and a lot more useful to a Bermuda captive reinsurer.

Critically, Bermuda captive reinsurers can pay more, can bid up the price, which is important in asset management circles (such as PE and private credit).

Guess what assets the Bermuda entities buy? Data center ABS, CLOs, High Yield corporate debt, and yes private software debt, the kind of stuff that costs too much capital to hold in size in the United States. These captive reinsurers are like expandable garbage bags that can take just about anything.

MetLife, a publicly traded insurer, keeps some of the data center ABS it creates for itself, and the rest goes to the balance sheet of its captive Bermuda reinsurer.

Not all life insurers are public. They can be private, and these days that means that they are likely owned by a PE firm.

Why would a PE firm want to own a life insurance company? PE firms often have less liquid investments that need a home. It can make investments, and the life insurance company can invest in the bonds and other securities backing the PE investment.

The PE firm can set up a captive reinsurer in Bermuda and feed the insurance policy risk onto its balance sheet, with lighter capital requirements. An expandable garbage bag that can take nearly any asset, exactly what the PE firm needs. The captive reinsurer therefore can buy more risky debt and securities, such as illiquid software company loans or data center ABS tranches.

Apollo is well known for its life insurer Athene, which between the parent and the captives are chock full of private credit loans for all manner of risky investments. Software loans, yes, and data center loans.

Bermuda has wonderful disclosure on this sort of thing. Easily discoverable details available to the enterprising analyst.

Of the 200 new Bermuda entities in the last few years, 36 were PE-affiliated captive reinsurers.

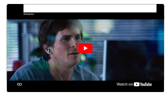
These PE firms are major backers of Private Credit funds that are creating and issuing data center ABS. And that is the purpose of the PE-affiliated captive reinsurers – to buy their own data center ABS.

Now, why would these U.S. public life insurance firms and PE-affiliated private life insurance firms suddenly go through the trouble to set up one hundred plus captive reinsurers?

Because data center ABS are a particularly risky new investment.

ABS are generally a pool of data center loans that is divided into tranches, which are rated by ratings agencies and sold at higher ratings than any loan would be sold at individually or even as an entire pool.

This is the exact structure of the residential mortgage-backed securitizations I shorted two decades ago. Perhaps you will recall the Margot Robbie scene in the movie.



The duration mismatch is the problem.

The question at the end of the day is whether those loans are good loans, and I just do not see a way that they are good. They are structured with the hyperscalers as sponsors who provide a loose backstop for the loans on their data centers. Sponsor support agreements, however are not strong backstops. There are outs for construction delays, power-delivery issues, and even short outages can trigger nonpayment of rent.

When Microsoft terminated some leases in 2025, it used the power-delivery reason for doing so. Meta did so as well.

More importantly, in a correlation event involving all AI data centers of a certain vintage, those backstops will not work. The backstop is called a Residual Value Guarantee. Moody's described the problem:

> An RVG acts as a backstop; if a tech company cancels or does not renew a lease, it must pay the landlord the difference if the data center's market value falls below a pre-agreed threshold.

In a correlation event, with all data centers of a vintage falling in market value together, the RVG backstop will have few teeth.

In most any situation that would cause that AAA, AA level hyperscaler to walk from a data center, there is a good likelihood a correlation event is happening.

Moody's found that the five major hyperscalers – Microsoft, Amazon, Alphabet, Meta, and Oracle – together have $662 billion off-balance sheet commitments already. GAAP accounting lets them hide it. Again, per Moody's:

> If a company concludes a lease renewal is likely to be exercised, but not **reasonably certain**, it can avoid classifying both the lease renewal periods and the residual value guarantee as liabilities.

A 60% certain renewal will be hidden. But they will all show up in a correlation event. And the hyperscalers will not play nice with the data center lenders.

If this goes bad, it might be at the hands of insurance regulators. In that case, there could be an unwind of a material but third tier funding source for AI data centers. Certainly, at the margin, data center projects might be abandoned, or sold. There is no telling what that may or may not trigger, given the AI sentiment at the time.

Another potential problem is that two of the big parent life insurance companies engaged in the Bermuda offshore business are also found holding some very large borrows from the Federal Home Loan Banks in Des Moines and New York.

This could be just the beginning of a new funding source for insurance companies' plans to invest in data center buildout investments, including power and commercial real estate related to the buildout.

Why the FHLB? It is a cheap funding source, and it is looking like there will be a lot of funding needed. The FHLB example here shows just how far the funding reach, the perimeter can stretch. Morgan Stanley estimates private credit from PE firms helped/funded by insurance companies will need to foot $800 billion of AI infrastructure buildout by 2028, about 1/3 of the total $2.9 trillion planned spending during that period.

If the need for data centers is less than has been planned, or data centers depreciate in economic terms much faster than the loans backing them, it may be up to Federal Home Loan Banks, insurance regulators and ratings agencies, including insurance ratings agencies, to call it when the first canary dies.

When a data center is first built, a fairly typical 2-3 year interest-only construction loan from a bank syndicate finances the build. Once the tenant (hyperscaler) takes over, permanent financing comes in to take out the construction loan.

This permanent financing can be the aforementioned asset-backed security, a privately placed bond, or a direct loan from an insurance company or private credit fund.

As discussed, the data center ABS market is small now – the Meta deal last year was almost the sum total. Insurance companies and PE firms with their private credit and insurance arms stand ready, if a little lame due to the software soft spot.

And they better be. Ropes & Gray's data center outlook for 2026 predicts there is a need for $275 billion in permanent loans to take out construction debt for data centers under construction right now. Over the next two years or so, either the ABS market needs to expand, or private credit (private equity, insurance) need to step up with direct loans.

There is an indication that traditional banks behind the construction loans are starting to make some judgments on how this is going to go. From Pere Credit on February 5th this year:

> Banks behind Oracle-backed construction projects are shopping loan stakes to commercial real estate private credit lenders and forcing deeper risk considerations as a consequence.

If one has stayed engaged this long, one would fully understand Pere Credit's warning.

This is the leading edge of the bezzle pausing, thinking, maybe even shrinking. Banks that initially wrote the Oracle/Stargate construction loans are now selling them down because the exposure is too concentrated in Oracle perhaps. The buyers are private credit funds, many of which are owned by or affiliated with the same PE houses that own those captive Bermuda entities.

Those private credit funds have been dealing with another problem, as discussed in Software & Payments Stocks Part 1. That is, they own too many illiquid private software company loans, and are starting to take markdowns on those. The PE firms also might own a software company or two in trouble.

This leaves an open question as to how healthy the private credit market will be when the time comes to take that ~$300 billion construction loan cliff challenge.

Now remember, thanks to the dynamics behind tokenmaxxing, there ought to be some concern about ultimate demand for these data centers, and future data centers. And there is that 19 GW power grid gap, as mentioned.

Time to close the loop and call it a day.

Tokenmaxxing is mostly rational but makes demand look bottomless.

This demand signal moves upstream through NVIDIA to TSMC's custom chip lines and downstream through too many construction loans needing permanent refinancing/takeout. So many loans of such size that insurers, private credit, captive reinsurers all must buy in to keep the bezzle alive.

The bezzle is not that AI is useless. The bezzle is that the market is capitalizing the most expensive phase of AI adoption as if it were normal and indicative of future demand. Benchmarking is not production, training is not performance. And tokenmaxxing is not a dependable infrastructure demand signal.

Compression is here, and is coming on harder. To reduce third-party token usage. To move up the pyramid, to substitution, to smaller models, cached context, and more efficient workflow routing.

So, yes, son, they are flying those airplanes around empty. The only time that made sense was at the beginning of flight. When pilots need hours, engineers need data and regulators need time. That benchmarking phase, all lusty fire and brimstone, needed at the very beginning of anything at all worthwhile.

They will not be doing it for long.

Until next time!

Cassandra Unchained is a reader-supported publication. To receive new posts as they are released and access the archives and Chat areas, please consider becoming a free or paid subscriber.

1,001 Likes · 108 Restacks

Discussion about this post

Comments Restacks

Write a comment...

Rebecca May 22
Liked by Michael Burry
Michael are you sleeping? Eating? How on earth did you manage this excellent article with the other 3, in depth posts, on deck. It's incredible, love it. Thanks so much job well done!
LIKE (38) REPLY SHARE

3 replies by Michael Burry and others

Lauren McDonald May 22
Liked by Michael Burry
I love the smell of 10qs in the am. This was so good thank you Mike!!!! Happy weekend I am watching this LIVE as a headhunter/recruiter. None of it makes sense and a lot of folks are looking to leave Nvidia. They are emailing me a ton.
LIKE (37) REPLY SHARE

1 reply

137 more comments...

© 2026 Michael Burry · Privacy · Terms · Collection notice
Substack is the home for great culture